

Digital Forensic Investigation Report

Forensic Analysis of the M57 Jean Disk Image

Case ID: DFIR-001

Case Name: M57 Jean Forensic Examination

Investigator: *Gabara Agata*

Organization: Personal Digital Forensics Portfolio

Tool Used: Autopsy 4.23.1

Evidence Source: M57 Jean Disk Image (Digital Corpora)

Evidence Format: EnCase E01

Report Date: 26 June 2026

Confidentiality Notice

This report was prepared for educational purposes using a publicly available forensic training image provided by Digital Corpora. All findings relate exclusively to the training dataset.

1.Executive Summary

The purpose of this investigation was to perform a forensic examination of the **M57 Jean** disk image using **Autopsy 4.23.1**. The analyzed evidence is part of the publicly available Digital Corpora forensic training dataset and was examined in a controlled laboratory environment.

The objective of the investigation was to identify and analyze digital artifacts that could be used to reconstruct user activity on the examined system. The analysis focused on operating system information, user accounts, installed software, browser artifacts, recently accessed documents, downloaded files, email artifacts, USB device history, and deleted files.

During the examination, multiple user accounts were identified, including the primary user account **jean**. The forensic analysis revealed evidence of web browsing activity, downloaded software, email-related artifacts, recently opened documents, and several connected USB devices. The operating system was identified as **Microsoft Windows XP Service Pack 3**, and the recovered artifacts provided valuable insight into the workstation's configuration and user activity.

All findings presented in this report were obtained using standard forensic procedures. The original evidence was examined in a read-only environment, ensuring that its integrity remained preserved throughout the investigation.

2. Case Background

The purpose of this investigation was to conduct a comprehensive forensic examination of the **M57 Jean** disk image using **Autopsy 4.23.1**. The evidence analyzed during this investigation is part of the publicly available **M57 Patents Scenario**, developed by Digital Corpora as a forensic training dataset.

The M57 dataset simulates a realistic corporate workstation environment and is widely used for digital forensic education and professional training. It contains a variety of operating system artifacts, user-generated content, application data, and system metadata that enable investigators to practice evidence identification, artifact correlation, and forensic reporting.

This investigation was performed in a controlled laboratory environment using a forensic copy of the original evidence. No modifications were made to the source image during the examination, ensuring that the integrity of the evidence was preserved throughout the analysis.

The examination focused on identifying artifacts that could assist in reconstructing user activity and understanding how the workstation was used. Particular attention was given to browser activity, user accounts, installed software, document access, removable storage devices, email artifacts, and deleted files.

3. Investigation Objectives

The primary objectives of this forensic examination were:

- Identify the operating system and workstation configuration.
- Enumerate all local user accounts.
- Examine installed software and applications.
- Analyze recently accessed documents.
- Review browser history, downloads, and bookmarks.
- Identify email-related artifacts.
- Detect removable USB storage devices connected to the system.
- Examine deleted files and recover available evidence.
- Review image metadata where available.
- Correlate recovered artifacts to reconstruct user activity.
- Document findings in accordance with digital forensic best practices.

4. Evidence Description

4.1 Evidence Source

The forensic examination was conducted using the **M57 Jean** training disk image obtained from the **Digital Corpora** repository. The dataset is intended for educational and research purposes and simulates a realistic Windows workstation used in a corporate environment.

The evidence was provided in the **EnCase Evidence File Format (E01)** and consisted of two segmented files:

Evidence ID	File Name	Format
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E-001	nps-2008-jean.E01	EnCase Evidence File
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E-002	nps-2008-jean.E02	EnCase Evidence File (Segment 2)
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The evidence image was imported directly into Autopsy for analysis without any modification to the original files.

4.2 Examination Environment

The forensic examination was performed in a controlled laboratory environment using the following software:

Component	Description
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Operating System	Windows 11
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Forensic Tool	Autopsy 4.23.1
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Evidence Format	EnCase (E01)
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Investigation Type	Static Disk Image Analysis
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The investigation was conducted using a read-only forensic image to ensure that the integrity of the original evidence was preserved throughout the examination.

4.3 Enabled Ingest Modules

The following Autopsy ingest modules were enabled during the examination:

- Recent Activity
- File Type Identification
- Extension Mismatch Detector
- Embedded File Extractor
- Picture Analyzer

- Email Parser
- Encryption Detection
- Interesting Files Identifier
- Keyword Search
- Hash Lookup

These modules were used to automatically identify and extract relevant forensic artifacts from the disk image.

5. Examination Findings

5.1 Operating System Information

The initial phase of the investigation focused on identifying the operating system and basic workstation configuration. System information recovered by Autopsy confirmed that the examined image originated from a Microsoft Windows XP workstation.

The recovered operating system details provide important contextual information and establish the foundation for further forensic analysis. Knowledge of the operating system version is essential for interpreting system artifacts, registry structures, file locations, and user activity.

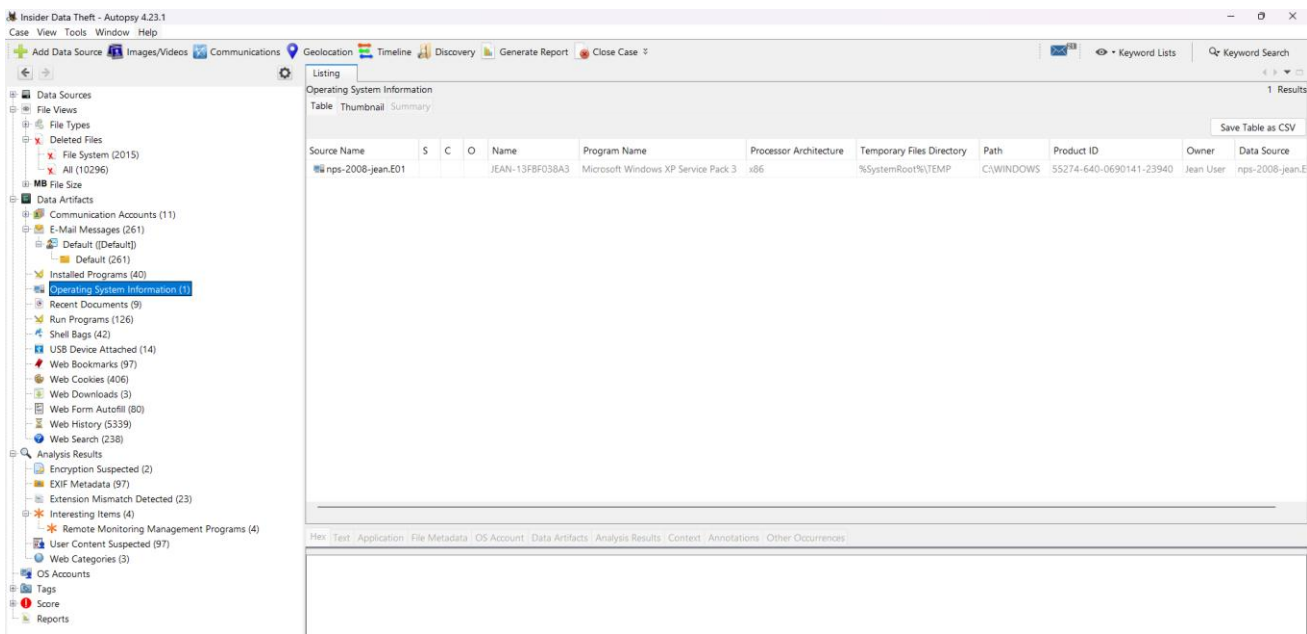
The following information was recovered during the examination:

Property	Value
Operating System	Microsoft Windows XP Service Pack 3
Architecture	x86 (32-bit)
Computer Name	JEAN-13FBF038A3
Windows Directory	C:\WINDOWS

The recovered operating system information confirms that the workstation was running Windows XP Service Pack 3. This information is consistent with the remaining forensic artifacts identified during the examination.

Evidence

Figure 2. Operating System Information



5.2 User Accounts

User account analysis was performed to identify individuals who had access to the examined workstation. Local user accounts provide valuable information during forensic investigations, as they help attribute system activity, document access, browser usage, and application execution to specific users.

Autopsy identified six local accounts within the examined Windows installation. These included two standard user accounts, one administrative account, and three default Windows service accounts.

The following accounts were identified:

Account	Description
Administrator	Built-in Windows administrative account.
jean	Primary user account associated with the workstation.

Account	Description
devon	Secondary local user account.
SYSTEM	Default Windows system account.
LOCAL SERVICE	Built-in service account used by Windows services.
NETWORK SERVICE	Built-in service account with limited network privileges.

Among the identified accounts, **jean** appears to be the primary workstation user. This conclusion is supported by the correlation of multiple forensic artifacts recovered during the examination, including recently accessed documents, browser activity, downloaded files, and email artifacts associated with this account.

The remaining built-in accounts (**SYSTEM**, **LOCAL SERVICE**, and **NETWORK SERVICE**) are standard Windows service accounts and do not indicate direct user interaction.

Identifying the primary user account is an important step in forensic investigations because it allows subsequent artifacts to be associated with a specific user profile and provides context for reconstructing user activity.

Evidence

Figure 3. Local User Accounts Identified by Autopsy

Name	S	C	O	Login Name	Host	Scope	Realm Name	Creation Time
S-1-5-21-484763869-796845957-839522			0	administrator	nps-2008-jean.E01_1 Host	Local		
S-1-5-21-484763869-796845957-839522			0	jean	nps-2008-jean.E01_1 Host	Local		
S-1-5-21-484763869-796845957-839522			0	devon	nps-2008-jean.E01_1 Host	Local		
S-1-5-19				LOCAL SERVICE	nps-2008-jean.E01_1 Host	Local	NT AUTHORITY	
S-1-5-20				NETWORK SERVICE	nps-2008-jean.E01_1 Host	Local	NT AUTHORITY	
S-1-5-18				SYSTEM	nps-2008-jean.E01_1 Host	Local	NT AUTHORITY	

5.3 Installed Programs

The analysis of installed software was conducted to identify applications available on the examined workstation. Installed applications provide valuable context regarding the system's intended use, user activities, and the types of artifacts that may be present during a forensic examination.

Autopsy recovered information indicating that the workstation contained a combination of operating system components, productivity software, web browsers, communication tools, and virtualization software.

The following applications were identified during the examination:

Application	Purpose
Mozilla Firefox	Web browser
Microsoft Office 2000	Office productivity suite
AIM Messenger	Instant messaging client
Adobe Flash Player	Browser multimedia plugin
VMware Tools	Virtual machine integration software

The presence of **Mozilla Firefox** is particularly significant, as it explains the browser-related artifacts recovered during the investigation, including browsing history, downloads, bookmarks, cookies, and cached web content.

The installation of **Microsoft Office 2000** indicates that the workstation was capable of creating and editing business documents such as spreadsheets, presentations, and word processing files. This finding is consistent with the recently accessed documents identified later in the investigation.

The presence of **AIM Messenger** suggests that the workstation was used for instant messaging. Although no conclusions regarding user communications can be drawn solely from installed software, messaging applications may contain valuable forensic artifacts depending on the available evidence.

The recovered software inventory provides important context for understanding the overall environment in which the examined workstation was operated and supports the interpretation of subsequent forensic artifacts.

Evidence

Figure 4. Installed Programs

Source Name	S	C	O	Program Name	Date/Time	Data Source
software	0			VMware Tools v.3.2.0.1288	2008-07-19 23:32:23 CEST	nps-2008-jean.E01
software	0			Microsoft Visual C++ 2005 Redistributable v.8...	2008-07-19 23:31:36 CEST	nps-2008-jean.E01
software	0			QQ Bubble Arena	2008-07-18 04:57:53 CEST	nps-2008-jean.E01
software	0			Aim Plugin for QQ Games	2008-07-18 04:31:42 CEST	nps-2008-jean.E01
software	0			QQ Games v.2.0.102.33	2008-07-18 04:31:41 CEST	nps-2008-jean.E01
software	0			iTunes	2008-07-18 04:30:49 CEST	nps-2008-jean.E01
software	0			AIM Toolbar 5.0 v.5.7.3.2	2008-07-18 04:29:28 CEST	nps-2008-jean.E01
software	0			Viewpoint Media Player	2008-07-18 04:29:26 CEST	nps-2008-jean.E01
software	0			AIM 6	2008-07-18 04:29:14 CEST	nps-2008-jean.E01
software	0			ACLOCP_Y	2008-07-18 04:28:51 CEST	nps-2008-jean.E01
software	0			ACLOCP_N	2008-07-18 04:28:48 CEST	nps-2008-jean.E01
software	0			WebFidrs XP v.9.50.7523	2008-07-12 03:05:46 CEST	nps-2008-jean.E01
software	0			MPlayer2	2008-07-12 03:05:29 CEST	nps-2008-jean.E01
software	0			Mozilla Firefox (3.0.1) v.3.0.1 (en-US)	2008-07-11 05:20:12 CEST	nps-2008-jean.E01
software	0			Adobe Flash Player ActiveX v.9.0.124.0	2008-07-10 20:09:03 CEST	nps-2008-jean.E01
software	0			ShockwaveFlash v.9.0.124.0	2008-07-10 20:09:03 CEST	nps-2008-jean.E01
software	0			Microsoft Office 2000 Premium v.9.00.2720	2008-07-06 07:38:43 CEST	nps-2008-jean.E01
software	0			Update for Windows XP (KB942763) v.1	2008-06-07 05:11:16 CEST	nps-2008-jean.E01
software	0			Security Update for Windows XP (KB923789)	2008-06-07 05:11:11 CEST	nps-2008-jean.E01
software	0			WIC	2008-05-14 07:20:43 CEST	nps-2008-jean.E01

5.4 Recent Documents

The **Recent Documents** artifact was examined to identify files that had been accessed by the primary user prior to the acquisition of the forensic image. This artifact is generated by the Windows operating system and records shortcuts to files that were recently opened by users. Although the presence of a shortcut does not confirm that a file was modified, it provides strong evidence that the document was accessed at a specific point in time.

The examination identified several recently accessed files associated with the **jean** user profile. The recovered artifacts included Microsoft Excel spreadsheets, image files, and Windows shortcut files.

Examples of recently accessed documents include:

File Name	File Type
m57biz.xls	Microsoft Excel Spreadsheet
tag-cloud.jpg	JPEG Image
LightBlueTop.gif	GIF Image
11soft.flipflops.jpg	JPEG Image

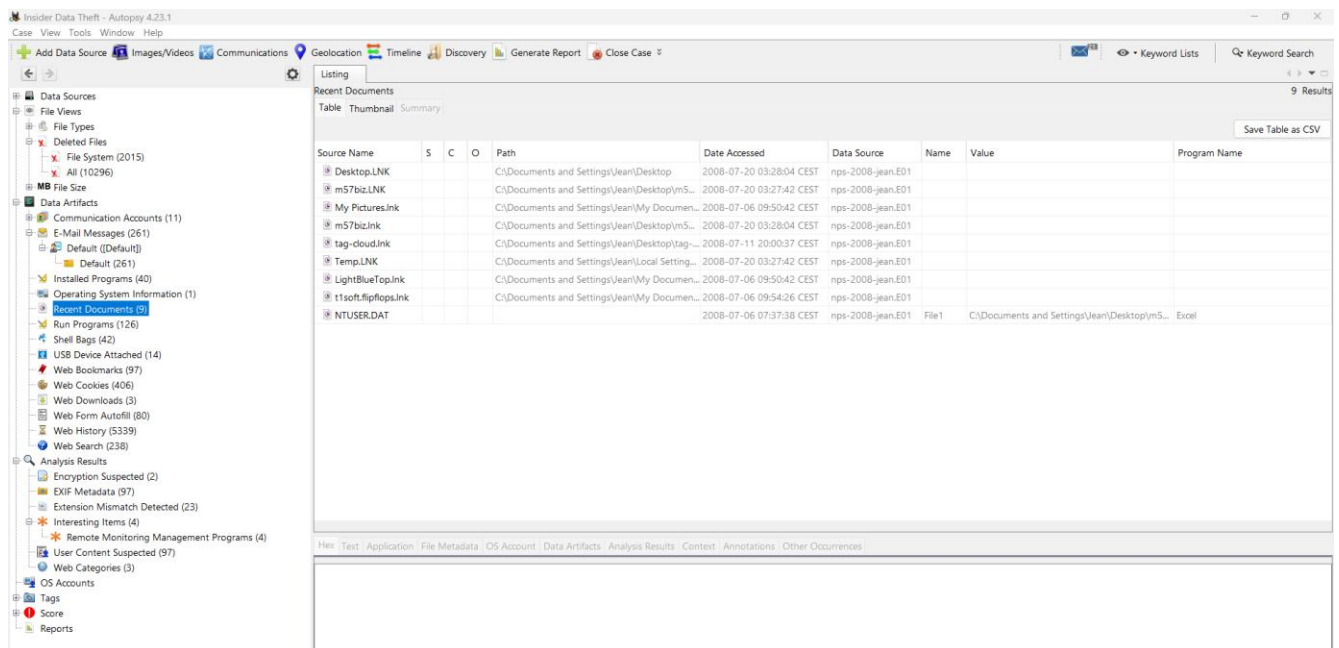
The presence of **m57biz.xls** indicates that spreadsheet documents were accessed on the workstation. Since Microsoft Office 2000 was identified during the software inventory, this finding is consistent with the installed applications and supports the conclusion that the workstation was used for office-related activities.

Several image files were also identified among the recently accessed documents. While these files do not independently indicate suspicious activity, they contribute to reconstructing normal user behavior and help establish a broader timeline of workstation usage.

Recent Documents artifacts are valuable because they enable investigators to correlate user activity with other evidence sources such as browser history, email artifacts, and file system metadata. When analyzed together, these artifacts provide a more complete picture of user interactions with the operating system.

Evidence

Figure 5. Recent Documents Recovered by Autopsy



5.5 Browser Activity

Browser artifacts were examined to reconstruct the user's online activity and identify websites accessed prior to the acquisition of the forensic image. Web browsing history is one of the most valuable sources of forensic evidence, as it provides insight into user behavior, search activity, downloaded content, and interactions with online services.

Autopsy recovered a substantial number of browser artifacts from the examined system, including browsing history, cookies, bookmarks, and cached web content. The recovered data indicates that **Mozilla Firefox** was the primary web browser used on the workstation, which is consistent with the installed software identified earlier in the investigation.

The browsing history contained over **5,300 recorded entries**, representing visits to various websites over an extended period. Among the recovered records were visits to search engines, software-related websites, news portals, and informational resources.

Examples of recovered browsing activity included:

- Google Search
- Mozilla websites
- CNN
- Centers for Disease Control and Prevention (CDC)

The browser history also contained search queries and page titles associated with publicly available information. These records provide evidence of websites accessed by the user but should always be interpreted within the broader context of the investigation. The presence of a visited webpage alone does not indicate malicious intent or unauthorized activity.

Browser artifacts are particularly valuable because they can be correlated with other forensic evidence such as downloaded files, recently accessed documents, email communications, and timeline events. This correlation enables investigators to reconstruct user activity with greater accuracy and establish relationships between different sources of evidence.

Evidence

Figure 6. Browser History Recovered by Autopsy

Source Name	S	C	O	URL	Date Accessed	Title	Program Name	Domain	Data Source	Ref
places.sqlite	1			http://search.cnn.com/search.jsp?query=anthrax...	2008-05-14 07:36:00 CEST	anthrax - Search results for anthrax - CNN.com	Firefox Analyzer	cnn.com	nps-2008-jean.E01	
places.sqlite	1			http://www.cnn.com/	2008-05-14 07:35:54 CEST	CNN.com - Breaking News, U.S. World, Weath...	Firefox Analyzer	cnn.com	nps-2008-jean.E01	
places.sqlite	1			http://download.mozilla.org/?product=firefox-...	2008-05-14 07:38:06 CEST	http://download.mozilla.org/?product=firefox-...	Firefox Analyzer	mozilla.org	nps-2008-jean.E01	
places.sqlite	1			http://minor.atlanticmetro.net/mozilla/firefox/...	2008-05-14 07:38:32 CEST	http://minor.atlanticmetro.net/mozilla/firefox/...	Firefox Analyzer	atlanticmetro.net	nps-2008-jean.E01	
places.sqlite	1			http://www.cdc.gov/sz/fhtml	2008-05-14 07:37:50 CEST	CDC A-Z Index - F	Firefox Analyzer	cdc.gov	nps-2008-jean.E01	
places.sqlite	1			http://www.google.com/	2008-05-14 07:37:50 CEST	Google	Firefox Analyzer	google.com	nps-2008-jean.E01	
places.sqlite	1			http://www.mozilla.com/en-US/firefox/all-beta...	2008-05-14 07:37:50 CEST	Firefox web browser International versions: G...	Firefox Analyzer	mozilla.com	nps-2008-jean.E01	
places.sqlite	1			http://www.cdc.gov/sz/a.html	2008-05-14 07:36:10 CEST	CDC A-Z Index - A	Firefox Analyzer	cdc.gov	nps-2008-jean.E01	
places.sqlite	1			http://www.cdc.gov/hcvdcd/dtdmd/disease_list...	2008-05-14 07:36:32 CEST	Disease Listing: Anthrax General Information [...]	Firefox Analyzer	cdc.gov	nps-2008-jean.E01	
places.sqlite	1			http://www.cdc.gov/sz/s.html	2008-05-14 07:37:54 CEST	CDC A-Z Index - S	Firefox Analyzer	cdc.gov	nps-2008-jean.E01	
places.sqlite	1			http://www.google.com/search?hl=en&q=firef...	2008-05-14 07:37:54 CEST	firefox 3 - Google Search	Firefox Analyzer	google.com	nps-2008-jean.E01	
places.sqlite	1			http://www.mozilla.com/firefox/all-beta.html	2008-05-14 07:37:56 CEST	http://www.mozilla.com/firefox/all-beta.html	Firefox Analyzer	mozilla.com	nps-2008-jean.E01	
places.sqlite	1			http://www.cdc.gov/sz/e.html	2008-05-14 07:36:28 CEST	CDC A-Z Index - E	Firefox Analyzer	cdc.gov	nps-2008-jean.E01	
places.sqlite	1			http://www.cdc.gov/hodod/dtdmd/diseaseinfo/...	2008-05-14 07:36:08 CEST	http://www.cdc.gov/hodod/dtdmd/diseaseinfo/...	Firefox Analyzer	cdc.gov	nps-2008-jean.E01	
places.sqlite	1			http://www.cdc.gov/sz/g.html	2008-05-14 07:36:28 CEST	http://www.cdc.gov/sz/g.html	Firefox Analyzer	cdc.gov	nps-2008-jean.E01	
places.sqlite	1			http://www.cdc.gov/od/oc/media	2008-05-14 07:36:22 CEST	http://www.cdc.gov/od/oc/media	Firefox Analyzer	cdc.gov	nps-2008-jean.E01	
places.sqlite	1			http://en-us.start2.mozilla.com/en-US/firefox/3...	2008-05-14 07:38:53 CEST	Welcome to Firefox 3 Beta 5	Firefox Analyzer	mozilla.com	nps-2008-jean.E01	
places.sqlite	1			http://en-us.start2.mozilla.com/firefox?client=...	2008-05-14 07:38:53 CEST	firefox	Firefox Analyzer	mozilla.com	nps-2008-jean.E01	
places.sqlite	1			http://www.google.com/firefox?client=firefox-a...	2008-05-14 07:38:50 CEST	Mozilla Firefox Start Page	Firefox Analyzer	google.com	nps-2008-jean.E01	

5.6 Downloaded Files

The analysis of browser download artifacts was performed to identify files obtained from the Internet and to better understand user activity on the examined workstation. Download records are valuable during forensic investigations because they provide evidence of software installations, document acquisition, and other files intentionally retrieved by the user.

Autopsy recovered multiple download records from the Mozilla Firefox browser. The recovered artifacts indicate that the user downloaded several software installation packages from trusted online sources.

Examples of recovered downloads include:

Downloaded File	Description
Firefox Setup 3.0 Beta 5.exe	Mozilla Firefox installer
install_aim.exe	AIM Instant Messenger installer
install_flash_player.exe	Adobe Flash Player installer

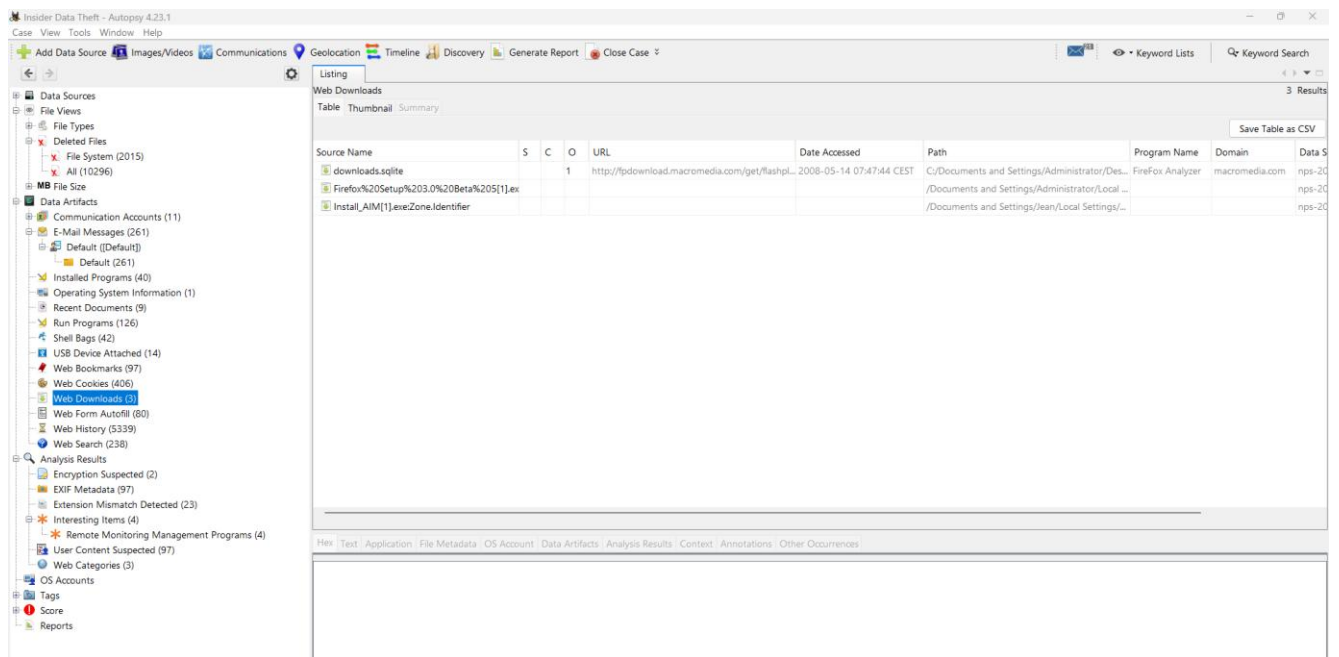
The recovered download history is consistent with the software inventory identified during the examination. Applications such as Mozilla Firefox, AIM Messenger, and Adobe Flash Player were present on the system, suggesting that at least some of the downloaded files were subsequently installed.

The correlation between browser download records and installed applications increases confidence in the forensic findings and helps reconstruct the sequence of user actions performed on the workstation.

No evidence of malicious software or unauthorized executable files was identified within the reviewed download records. However, download artifacts remain an important source of evidence, as they document user-initiated file acquisition and may assist investigators in reconstructing system activity.

Evidence

Figure 7. Browser Download History



5.7 Email Artifacts

Email artifacts were examined to identify evidence of electronic communications stored within the forensic image. Email data is frequently considered a valuable source of digital evidence, as it may provide information regarding user interactions, document exchange, account activity, and communication timelines.

Autopsy identified a total of **261 email artifacts** associated with the examined workstation. The recovered data indicates that email-related information was successfully extracted from the forensic image and indexed for further analysis.

Although the objective of this examination was not to perform a detailed content analysis of individual messages, the recovered artifacts confirm that email activity formed part of the user's normal workstation usage. The existence of these records provides investigators with an additional source of evidence that can be correlated with browser history, downloaded files, recently accessed documents, and timeline events.

Email artifacts are particularly valuable because they preserve metadata such as message timestamps, sender and recipient information, subject lines, and attachment references. When examined alongside other forensic artifacts, this information can assist in reconstructing user activity and establishing relationships between events.

During this investigation, the identified email artifacts were documented as available evidence for further examination if required. No conclusions regarding the content or intent of individual communications were made without additional analysis.

Evidence

Figure 8. Email Artifacts Identified by Autopsy

Source Name	S	C	O	E-Mail From	Subject	Date Received	Message (Plaintext)	Message ID	Has Att
outlook.pst				Microsoft Outlook 2000	Welcome to Microsoft Outlook 2000!	2008-07-10 09:47:23 CEST	Welcome to Microsoft Outlook 2000 One Win...	2097188	Yes
outlook.pst				Google Alerts <googlealerts-noreply@google...	Google Alert - skin in the office	2008-07-18 05:19:43 CEST		2103492	
outlook.pst				Google Alerts <googlealerts-noreply@google...	Google Alert - m57.biz	2008-07-16 20:55:25 CEST		2103012	
outlook.pst				alex <alex@m57.biz>	FW: UFOs Over U.S. Military Sites?	2008-07-20 01:32:54 CEST	CNN.com - Larry King-----Original Message...	2104612	
outlook.pst				alex <alex@m57.biz>	FW: Making People Sick AND Poor	2008-07-20 01:32:54 CEST	CNN.com - Lou Dobbs Tonight-----Original M...	2104644	
outlook.pst				alex <alex@m57.biz>	FW: Subject line: Missing girl's mom borrowed ...	2008-07-20 01:32:54 CEST	CNN.com - Prime News-----Original Message...	2104676	
outlook.pst				alex <alex@m57.biz>	FW: The CNN Political Ticker AM for Friday, Jul...	2008-07-20 01:32:52 CEST	CNN.com - CNN Political Ticker-----Original M...	2104580	
outlook.pst				alex <alex@m57.biz>	FW: Fans ready to stay up all 'Knight' for Batm...	2008-07-20 01:32:52 CEST	-----Original Message-----From: CNN AM Qui...	2104516	
outlook.pst				alex <alex@m57.biz>	FW: All In All, I Feel Like Another Brick In the ...	2008-07-20 01:32:52 CEST	The Morning Express with Robin Meade-----O...	2104548	
outlook.pst				alex <alex@m57.biz>	RE: which email address are you using?	2008-07-20 01:50:19 CEST	Yes, I got this email-----Original Message-----	2105508	
outlook.pst				Microsoft Outlook 2000	Welcome to Microsoft Outlook 2000!	2008-07-06 09:38:43 CEST	Welcome to Microsoft Outlook 2000 One Win...	2097188	Yes
outlook.pst				Jean User <jean@m57.biz>	test test test	2008-07-06 09:48:26 CEST	Do I have email now?	2097380	
outlook.pst				Jean User <jean@m57.biz>	let's try again	2008-07-06 09:48:26 CEST	Do I suck?	2097412	
outlook.pst				jean@m57.biz <jean@m57.biz>	this is what I was talking about	2008-07-06 09:55:47 CEST	*Please note, the sender's email address has n...	2097508	
outlook.pst				Google Alerts <googlealerts-noreply@google...	Click to confirm your Google Alert	2008-07-06 09:56:41 CEST	Google received a request to start sending Ale...	2097540	
outlook.pst				Google Alerts <googlealerts-noreply@google...	Click to confirm your Google Alert	2008-07-06 09:56:53 CEST	Google received a request to start sending Ale...	2097572	
outlook.pst				accounts-noreply@google.com <accounts-nor...	Google Email Verification	2008-07-06 09:58:25 CEST	Welcome to Google Accounts. To activate you...	2097604	
outlook.pst				NPR Accounts <npr-accounts@npr.org>	Activate Your Account	2008-07-06 09:59:49 CEST	Thank you for registering for an account with ...	2097636	
outlook.pst				Google Alerts <googlealerts-noreply@google...	Google Alert - skin in the office	2008-07-06 10:00:35 CEST		2097700	

5.8 USB Device Analysis

The examination included an analysis of USB device artifacts to identify external storage devices that had been connected to the examined workstation. USB forensic artifacts are particularly important because removable media is frequently used to transfer, copy, or exfiltrate data.

Autopsy recovered records indicating that several USB devices had been connected to the system during its operational lifetime. The identified devices included both removable storage media and peripheral hardware.

The following devices were identified during the examination:

Device	Description
Apple iPhone	Mobile device
Flash Disk 256 MB	Removable USB storage device
CBM2080 / CBM2090 Flash Drive Controller	USB flash drive controller
VMware Virtual USB Hub	Virtual USB device
VMware Virtual HID Mouse	Virtual input device

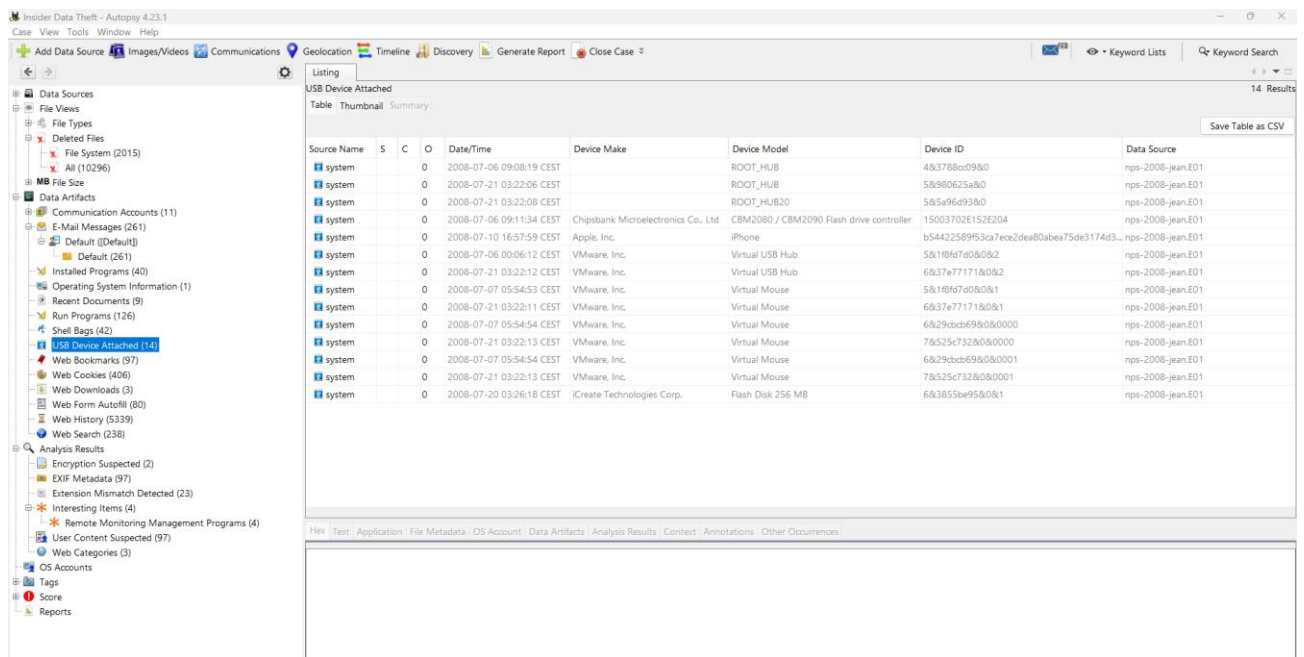
The presence of removable USB storage devices confirms that the workstation interacted with external media. While the recovered artifacts demonstrate that these devices were connected to the system, they do not independently establish that files were copied or transferred. Additional analysis of file system metadata, shellbags, link files, registry artifacts, or timeline data would be required to determine whether data movement occurred.

The identification of an Apple iPhone further indicates that a mobile device was connected to the workstation. Such connections may be associated with charging, synchronization, backup operations, or file transfers; however, the available artifacts do not provide sufficient evidence to determine the specific purpose of the connection.

From a forensic perspective, USB device artifacts provide valuable contextual information and can support investigations involving data exfiltration, intellectual property theft, or unauthorized use of removable media.

Evidence

Figure 9. USB Device History Recovered by Autopsy



5.9 Deleted Files

The examination included an analysis of deleted file artifacts to determine whether recoverable evidence remained on the disk image. Deleted files are of particular forensic interest because removing a file through the operating system does not immediately erase its contents from the storage media. Instead, the file system typically marks the associated disk space as available for reuse, allowing forensic tools to recover deleted data until it is overwritten.

Autopsy identified deleted file artifacts within the examined image and categorized them separately from active files. These artifacts were reviewed to determine whether they contained information relevant to the investigation.

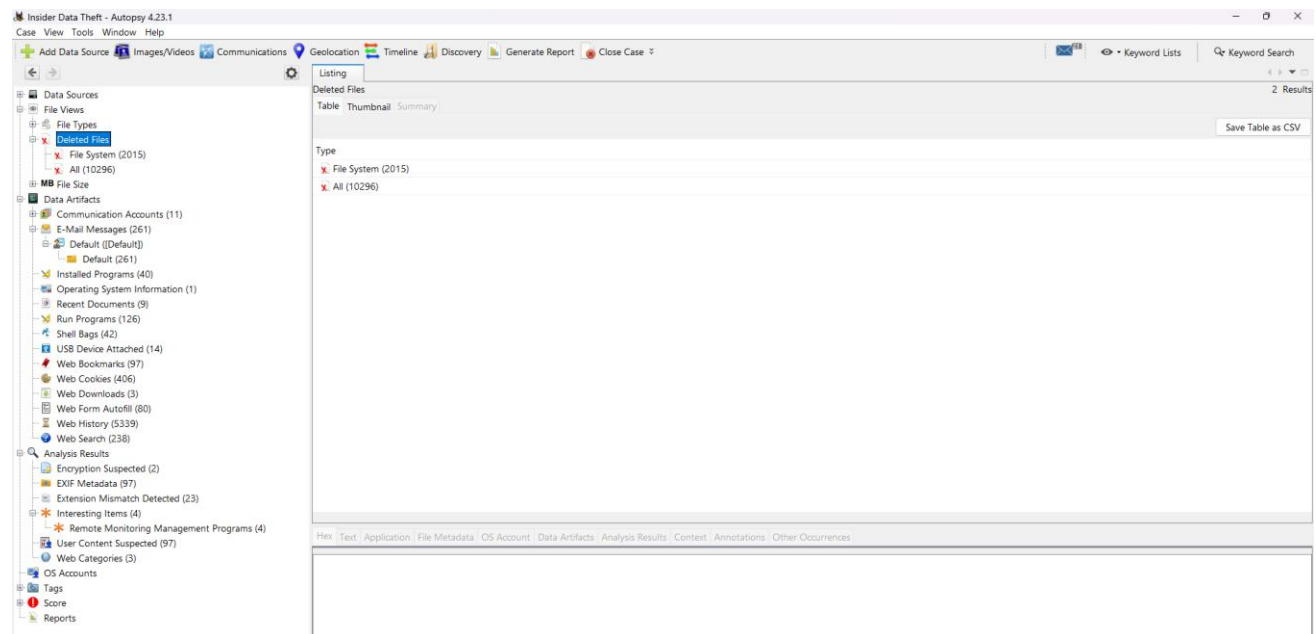
The presence of recoverable deleted files demonstrates that remnants of previously stored data remained accessible despite user deletion. Such artifacts may provide valuable insight into historical user activity, document handling, and file lifecycle events.

Deleted file analysis is an important component of digital forensic investigations because it can reveal evidence that users intentionally or unintentionally attempted to remove. When correlated with other artifacts, such as browser history, recently accessed documents, and timeline events, deleted files contribute to a more complete reconstruction of workstation activity.

During this examination, deleted file artifacts were documented and preserved as part of the overall evidence set. No evidence was modified or restored to the original storage media during the recovery process.

Evidence

Figure 10. Deleted Files Identified by Autopsy



5.10 Timeline Analysis

Timeline analysis was performed to reconstruct the chronological sequence of events recorded within the forensic image. By correlating timestamps from multiple artifact sources, investigators can establish when specific activities occurred and identify relationships between user actions and system events.

Autopsy automatically generated a timeline by combining metadata from the file system, browser artifacts, recently accessed documents, email records, and other operating system artifacts. This consolidated view enables investigators to analyze user activity in chronological order rather than examining individual artifacts in isolation.

The timeline provides a structured overview of workstation usage and assists in identifying periods of increased user activity, document access, software execution, browser sessions, and

file operations. Correlating these events with additional forensic artifacts improves the reliability of investigative findings and supports the reconstruction of user behavior.

Although the timeline identified numerous events throughout the lifetime of the examined system, no conclusions were drawn based solely on timestamp information. Timeline analysis was used in conjunction with supporting artifacts to ensure that findings were based on multiple independent sources of evidence.

The chronological reconstruction of events demonstrated the effectiveness of timeline analysis as a method for understanding system usage and validating relationships between recovered forensic artifacts.

Evidence

Figure 11. Timeline Analysis Generated by Autopsy



6. Conclusion

The forensic examination of the **M57 Jean** disk image was successfully completed using **Autopsy 4.23.1**. The investigation achieved its primary objective of identifying and

documenting digital artifacts that provide insight into the configuration of the examined system and the activities performed by its users.

Throughout the analysis, a wide range of forensic artifacts was recovered and examined, including operating system information, local user accounts, installed software, recently accessed documents, browser history, downloaded files, email artifacts, USB device records, and deleted files. The correlation of these artifacts enabled the reconstruction of user activity and provided a comprehensive understanding of how the workstation was used.

The examination demonstrated the value of forensic artifact correlation. Individual artifacts, while informative on their own, become significantly more valuable when analyzed together. Browser history, recent document records, installed applications, and removable device artifacts collectively contributed to a more accurate reconstruction of workstation usage.

No modifications were made to the original evidence during the examination. All analysis was performed on a forensic image within a controlled laboratory environment, following fundamental principles of digital forensic methodology and evidence preservation.

This investigation demonstrates the effectiveness of Autopsy as a digital forensic platform for the identification, recovery, and analysis of artifacts stored within Windows-based systems. The recovered evidence illustrates how forensic tools can assist investigators in reconstructing user activity while maintaining the integrity of digital evidence.

Although this examination was conducted using a publicly available training dataset, the investigative methodology and analytical process reflect techniques commonly applied during real-world digital forensic investigations.

7. Limitations

The findings presented in this report are based exclusively on the forensic artifacts recovered from the **M57 Jean** disk image. No live system acquisition, volatile memory analysis, network traffic analysis, or external log sources were available during the examination.

The investigation was limited to the artifacts successfully extracted by Autopsy from the acquired evidence image. Consequently, activities that did not leave persistent artifacts on the storage media could not be evaluated.

Additionally, the absence of a forensic artifact should not be interpreted as evidence that a particular activity did not occur. Digital artifacts may be altered, deleted, overwritten, or unavailable due to normal operating system behavior.

All conclusions presented in this report are supported by the evidence available at the time of the examination.

8. Recommendations

Based on the findings of this examination, the following additional forensic activities are recommended to further enhance the investigation:

- Perform Windows Registry analysis to identify additional user activity and system configuration changes.
- Analyze Windows Event Logs to reconstruct authentication events and system activity.
- Conduct memory analysis using Volatility to identify running processes, injected code, and network connections.
- Perform hash verification of recovered files using known hash databases.
- Examine shellbags, jump lists, prefetch files, and link files for additional user activity.
- Correlate forensic artifacts with external log sources where available.
- Preserve all recovered evidence in accordance with digital evidence handling procedures.

These additional activities would provide greater visibility into system usage and further strengthen the overall forensic investigation.

9. References

1. Carrier, B. *File System Forensic Analysis*. Addison-Wesley, 2005.
2. Casey, E. *Digital Evidence and Computer Crime*. Academic Press, 2011.
3. NIST Special Publication 800-86 – *Guide to Integrating Forensic Techniques into Incident Response*.

4. Autopsy User Documentation.
5. Digital Corpora – M57 Training Dataset.
6. NIST Computer Forensics Tool Testing Program (CFTT).

10. Appendix

Appendix A – Screenshots

Figure	Description
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Figure 1	Autopsy Case Overview
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Figure 2	Operating System Information
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Figure 3	User Accounts
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Figure 4	Installed Programs
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Figure 5	Recent Documents
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Figure 6	Browser History
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Figure 7	Browser Downloads
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Figure 8	Email Artifacts
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Figure 9	USB Device History
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Figure 10	Deleted Files
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Figure 11	Timeline Analysis
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Appendix B – Tool Information

Component	Version
Autopsy	4.23.1

Component	Version
Operating System	Windows 11
Evidence Format	EnCase E01
Dataset	Digital Corpora – M57 Jean